

MECH191 — Metallurgy

Week 8

Non-Destructive Testing Methods

October 12, 2026 · 5:00 – 7:30 PM · Kai Santos

Class Agenda

01

Surface & Near-Surface Methods

Visual inspection, dye penetrant, and magnetic particle — what each finds and where each falls short

02

Volumetric Methods

Ultrasonic and radiographic testing — detecting internal defects, technique selection, and safety

03

Method Selection & Qualification

Matching the method to the defect, written procedures, and ASNT certification levels

Module 7 • Week 8

NDT Methods

What to do this week:

Research 1018 steel, 360 brass, and Ti-6Al-4V: typical inclusions, intentional particles, and why NDT replaces sectioning here.

Steps:

- 1 Research — record data & sources
- 2 Answer — short-answer questions
- 3 Apply — critical-thinking & equations

Alloys of Record

- Alloy A — 1018 steel
- Alloy L — 360 leaded brass
- Alloy P — Ti-6Al-4V titanium

How to Complete This Module:

- 1 Research the alloy(s) using credible sources
- 2 Record data with a source citation
- 3 Answer the short-answer questions
- 4 Answer the critical-thinking questions
- 5 Keep the module with your portfolio

MECH 191 — Course & Unit Roadmap

Unit 1 Materials Fundamentals	Unit 2 Steel & Alloys	Unit 3 Testing & Quality	Unit 4 Manufacturing	Unit 5 Joining & Failure
Wk 1 Aug 24 Atomic Structure & Bonding	Wk 4 Sep 14 Carbon & Alloy Steels	Wk 7 Oct 5 Mechanical Testing	Wk 10 Oct 26 Casting	Wk 14 Nov 23 Welding Processes
Wk 2 Aug 31 Mechanical Properties	Wk 5 Sep 21 Stainless & Cast Irons	Wk 8 ← here Oct 12 NDT Methods	Wk 11 Nov 2 Forming	Wk 15 Nov 30 Weld Metallurgy & Failure
	Wk 6 Sep 28 Nonferrous & Phase Diagrams	Wk 9 Oct 19 Quality Standards	Wk 12 Nov 9 Machining	Wk 16 Dec 7 Corrosion & Wrap-up
			Wk 13 Nov 16 Powder Metallurgy	

Week 17 • FINAL EXAM

Week 8 — At a Glance

Topics

- Visual inspection (VT) — the first step in every inspection sequence; capabilities and limits
- Dye penetrant testing (PT) — capillary action, color contrast vs. fluorescent, dwell time, and surface limitations
- Magnetic particle testing (MT) — ferromagnetic materials only; yoke vs. prod; field orientation and 2-direction rule
- Ultrasonic testing (UT) — sound waves, couplant, straight beam vs. angle beam, A-scan, phased array
- Radiographic testing (RT) — X-ray vs. gamma sources, what volumetric defects look like, planar defect limitation
- Method selection — matching defect type, material, and geometry to the right method
- Written procedures and ASNT SNT-TC-1A qualification levels (I, II, III)

Resources

Reference Material

Moniz Ch. 29
Nondestructive Examination

Kalpakjian Ch. 4
Surfaces, Inspection, & Product Quality

Reference Pages

NDT Methods — Physics, Techniques & Limitations
Defect Recognition

Unit 3 Quiz — Weeks 7-9

Scheduled during Week 9

Core Questions

Try to answer these before we dive in — we'll revisit them with answers at the end of class.

1

What is the difference between destructive and non-destructive testing, and why does it matter in practice?

2

What are the five main NDT methods and what types of defects can each detect?

3

How do you choose the right NDT method for a given situation — what factors must you consider?

4

What role do written procedures and technician qualification play in NDT — and why are they non-negotiable?

Visual Inspection (VT) — The Foundation of Every Inspection

What VT Can and Cannot Find

Direct VT:

Naked eye + adequate lighting, mirrors, and magnifying lenses — the simplest and most universal method. Minimum illuminance is specified in the procedure.

Remote VT:

Borescopes, videoscopes, and drones reach where direct access is impossible — internal pipe surfaces, turbine blades, vessel interiors.

What VT finds:

Surface cracks, corrosion, mechanical damage, weld surface defects (undercut, overlap, spatter, surface porosity), incorrect geometry, misalignment, coating condition.

What VT misses:

Anything subsurface — a visually clean surface gives zero information about what lies beneath. VT is always Step 1; it is never the only step for critical components.

Weld Inspection by VT

- Check weld profile against the weld symbol on the drawing — correct size, length, and location
- Undercut: a groove at the weld toe that reduces effective throat thickness — measured with a fillet gauge
- Overlap: weld metal rolled over base metal without fusion — visible as a rolled edge; planar defect, no acceptable amount
- Excessive spatter: indicates incorrect parameters; may also indicate moisture in consumables
- Surface porosity: visible pores in the weld face — investigate shielding gas coverage and base metal cleanliness
- Cracks: linear indications at the weld surface, HAZ, or crater — any crack is cause for rejection pending investigation
- Key requirement: lighting ≥ 500 lux (or per procedure) — poor lighting is the most common cause of missed surface defects

Dye Penetrant Testing (PT) — How It Works and Where It Fails

Process Steps

- 1 Pre-cleaning:**
Remove all paint, scale, oil, and coatings — any contamination can seal the defect opening and prevent penetrant entry
- 2 Penetrant application:**
Apply to the cleaned surface; low-viscosity liquid is drawn into defects by capillary action
- 3 Dwell time:**
Hold time specified in the procedure — too short misses fine defects; too long may dry the penetrant in the defect before developer is applied
- 4 Excess removal:**
Remove surface penetrant without flushing the defect — use a damp cloth (solvent removable) or water rinse (water-washable)
- 5 Developer application:**
Draws trapped penetrant back out of defects — creates a visible bleed-out indication against a white background
- 6 Inspection + post-clean:**
Inspect per procedure; document indications; clean the surface to remove developer and residual penetrant

Color Contrast vs. Fluorescent

Color contrast (visible):

Red penetrant on white developer — inspected under white light. Standard field method, simpler logistics.

Fluorescent:

Penetrant glows under UV (black) light — significantly more sensitive; used for critical or fine-defect applications.

Limitations

- Only detects surface-breaking defects — subsurface defects are invisible to PT
- Surface must be clean and free of coatings that seal the defect opening
- Porous materials (as-cast surfaces, thermal spray coatings) cannot be inspected — the whole surface bleeds back
- Works on any non-porous material — metals, ceramics, plastics (unlike MT which needs ferromagnetic material)

Magnetic Particle Testing (MT) — Principle, Methods & Limitations

Principle & Magnetization Methods

How it works:

The component is magnetised; at a defect, the magnetic field is forced out of the material creating flux leakage; iron particles (dry or wet suspension) are attracted to the leakage and form a visible indication.

Yoke:

Electromagnetic yoke placed on the surface induces a field between its poles — portable, widely used for weld inspection; detects defects perpendicular to the field.

Prod:

Electrical contacts pressed onto the surface pass current through the material — creates a circular field; detects defects running between the prods; risk of arc burns at contact points.

Coil / central conductor:

Used for magnetising whole components longitudinally or circularly — common in shop environments for valves, fittings, and bar stock.

Key Rules & Limitations

Ferromagnetic only:

Carbon steels, low-alloy steels, martensitic and ferritic stainless steels. Cannot be used on austenitic stainless, aluminum, copper, or titanium.

Depth capability:

Detects surface and near-surface defects up to approximately 2–3 mm below the surface — superior to PT for near-surface detection in magnetic materials.

Two-direction rule:

A defect must be at 45–90° to the field to produce a detectable indication. Inspections must be performed in at least two perpendicular directions to ensure all orientations are covered.

WFMT:

Wet fluorescent MT: iron particles in liquid carrier viewed under UV light — the most sensitive MT technique; widely used for critical weld inspection in pressure vessel and pipeline fabrication.

Demagnetization:

Required after inspection on components where residual magnetism would interfere with service — near instruments, or where particles could contaminate a process.

Ultrasonic Testing (UT) — Detecting Internal Defects with Sound

Principle & Techniques

How it works:

A transducer converts electrical energy to high-frequency sound waves (1–10 MHz) that travel through the material. Reflections from defects or the back wall return to the transducer; time of flight gives depth.

Couplant:

Gel, oil, or water between transducer and surface — air is a poor transmitter of ultrasound. Couplant must be compatible with the material and the process.

Straight beam (compression wave):

Transducer perpendicular to the surface — sound travels straight through. Used for lamination detection and thickness measurement in corroded pipe.

Angle beam (shear wave):

Transducer angled to direct sound at 45°, 60°, or 70° into the material — used for weld inspection where the sound must be directed toward fusion faces and the weld root.

Reading Results & Advanced Techniques

A-scan display:

Time-base trace showing signal amplitude vs. time of flight — the inspector reads peaks (echoes) to identify defect depth and relative size. Fundamental UT display.

Phased array UT (PAUT):

Multiple elements steer and focus the beam electronically — produces S-scans (sectoral cross-sections of the weld). Increasingly replaces conventional UT and radiography in modern fabrication.

Thickness measurement:

Straight beam UT from one side gives remaining wall thickness — essential for in-service corrosion monitoring where internal access is impossible.

Limitations:

Rough surfaces reduce coupling efficiency. Coarse-grained materials (some austenitic stainless, castings) scatter sound and make interpretation difficult. Defects parallel to the beam may not reflect sufficient energy.

Radiographic Testing (RT) — Volumetric Imaging with X-Ray and Gamma

Principle & Sources

How it works:

Radiation passes through the component onto film or a digital detector on the opposite side. Denser material absorbs more radiation and appears lighter on the image; voids absorb less and appear darker.

X-ray sources:

X-ray tubes generate radiation with adjustable energy — suitable for a wide range of thicknesses; require electrical power. Used in shop environments.

Gamma ray sources:

Radioactive isotopes (Ir-192, Co-60, Se-75) — compact, no power required, fixed energy output. Ideal for field use and confined spaces where power is unavailable.

Film RT vs. digital:

Film provides a permanent archival image processed in a darkroom. Digital radiography (DR) and computed radiography (CR) use digital detectors or phosphor plates — faster, no chemicals, images can be enhanced.

What RT Finds, Misses, and Safety

Detects well:

Volumetric defects — porosity, slag inclusions, tungsten inclusions, burn-through, lack of fill. The rounded or irregular shape creates a clear density difference on film.

Detects poorly:

Tight planar defects (cracks, lack of fusion) oriented at a small angle to the beam may be completely invisible on a radiograph. UT is superior for planar defect detection.

Permanent record:

RT produces an image record that can be stored, reviewed, and compared over time — valuable for in-service monitoring and fitness-for-service assessments.

Radiation safety:

RT involves ionising radiation subject to strict regulatory controls — designated radiation areas, dosimetry monitoring, and licensing. A technician working near RT must understand exclusion zone requirements and never enter a restricted area during an exposure.

Method Selection — Matching the Method to the Problem

Ask five questions in sequence — the answers eliminate unsuitable methods and point to the correct choice.

Factor	VT	PT	MT	UT	RT
Defect location	Surface only	Surface-breaking only	Surface & near-surface	Surface & deep internal	Surface & deep internal
Material type	Any	Non-porous any material	Ferromagnetic only	Most metals & composites	Most solid materials
Detects planar cracks well?	Depends (surface)	Yes (if surface-breaking)	Yes (near-surface)	Yes (preferred)	Poor (if parallel to beam)
Access required	One side (direct view)	One side	One side	One side	Both sides (source + film)
Produces permanent image record?	No	No	No	A-scan / S-scan digital	Yes (film or digital)

ASNT SNT-TC-1A — Level I: perform under supervision • Level II: perform, interpret & accept/reject • Level III: write procedures & certify

Core Questions — Answers

What is the difference between destructive and non-destructive testing, and why does it matter?

Destructive testing (tensile, hardness, Charpy) consumes or damages the specimen — accurate but the tested piece cannot go into service. NDT inspects the actual component without altering it, enabling 100% inspection and in-service monitoring of critical components.

What are the five main NDT methods and what does each detect?

VT: visible surface defects. PT: surface-breaking defects in any non-porous material. MT: surface and near-surface defects in ferromagnetic materials. UT: surface and deep internal defects via sound waves. RT: volumetric defects (porosity, inclusions) via radiation imaging.

How do you choose the right NDT method?

Consider: (1) defect type and expected location (surface, near-surface, volumetric); (2) material (ferromagnetic or not, porous or not); (3) geometry and access available; (4) required sensitivity; (5) applicable code or standard specifying the method.

What role do written procedures and technician qualification play — and why are they non-negotiable?

Every method must be performed per a qualified written procedure specifying equipment, materials, technique, and acceptance criteria. Technicians must hold ASNT Level I, II, or III certification for the method. An uncertified person cannot accept or reject an indication, and deviating from the procedure invalidates the inspection regardless of the result.

Key Takeaways

1

Each NDT method operates on a different physical principle and detects a different subset of defects — no single method finds everything, and method selection must be matched to the defect type, material, geometry, and applicable code requirement.

2

Written procedures and technician qualification are not bureaucratic formalities — they are what makes NDT results legally and technically defensible. An inspection performed outside its qualified procedure is not a valid inspection regardless of what the technician found.

3

Understanding the limitations of each method is just as important as understanding its capabilities — knowing what a method cannot detect is what prevents a false sense of security after an inspection passes.

Resources & What's Next

Week 8 Resources

Moniz Ch. 29

Nondestructive Examination — primary reading

Kalpakjian Ch. 4

Surfaces, Inspection, and Product Quality Assurance

NDT Methods Reference

Physics, techniques, and limitations for all 5 methods

Defect Recognition Guide

Visual identification of common weld and material defects

Week 9 — Quality Standards

October 19, 2026

- What quality standards are and why they exist — ASTM, AWS, ASME, ISO, and API
- Common weld defects and how to recognize them — porosity, LOF, cracks, undercut
- Indication vs. discontinuity vs. defect vs. reject — precise language that matters
- What good inspection documentation looks like and why it is a legal document